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SWE1015- BIOMETRIC SYSTEMS

Embedded Project J-Component

REVIEW – 3

**TOPIC: FACE MASK DETECTION AND RECOGNITION SYSTEM
USING MACHINE LEARNING**

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ABSTRACT

The Coronavirus epidemic has had a profound effect on human health and well-being which requires the creation of a safe and secure environment. The COVID-19 virus is most prevalent in infected people who come in close contact with other people. So the only solution for preventing physical contact is by using face masks. The Face Mask will act as a physical barrier between you and these respiratory drops. In this paper, we have developed a face mask detection system that determines whether a person wears a mask or not and alerts him/her with an alarm sound. Additionally, the system can identify the person's name even if the mask is worn. The warning system will be integrated with the face mask detection system in three stages. The first stage involves training the dataset using the convolutional neural network "MobileNetV2". The second stage involves verifying whether a person is wearing a mask or not and identifying the person using face recognition. A warning system is developed in stage three. When a person is found not wearing a mask, he/she will be warned with an alarm. We developed an efficient and user-friendly face mask detection system, enhanced with name recognition, which can be used in many shops, schools, etc.

Keywords: COVID-19, facenet , alarm, mask, convolutional neural network, face recognition.

INTRODUCTION

The COVID-19 pandemic has significantly affected daily life, highlighting the need for effective safety measures to protect public health. The virus primarily spreads through respiratory droplets when individuals are in close proximity, making face masks an essential tool for minimizing physical contact and reducing transmission risks. In light of this, we have developed a face mask detection system that goes beyond simply identifying mask compliance. This system can also recognize the identity of a person even when they are wearing a mask. Such a feature is particularly valuable in settings where personal identification is critical, such as educational institutions, workplaces, and public venues, while still adhering to health and safety guidelines.

The proposed system employs the convolutional neural network "MobileNetV2" to efficiently determine whether a person is wearing a mask. Additionally, it incorporates a face recognition module capable of identifying individuals even when their facial features are partially obscured by the mask. If a person is not wearing a mask, the system generates an alarm as a warning. This system offers a dual benefit: ensuring compliance with mask mandates while maintaining reliable personal identification. By combining mask detection and masked face

recognition, this innovative solution provides a practical and effective approach to managing safety protocols in diverse environments.

LITERATURE SURVEY

Title	Authors	Year	Summary	Pros	Cons
Face Mask Detection Model	Goyal et al.	2022	Developed a face mask detection model for static and real-time videos, classifying images as "with mask" or "without mask." Achieved 98% accuracy using a Kaggle dataset of ~4,000 images.	High accuracy, computationally efficient, suitable for real-time use.	Limited dataset size compared to more recent studies.
Improved Tiny YOLO v4-SPP for Mask Detection	Kumar et al.	2023	Proposed a novel face mask detection system using Tiny YOLO v4-SPP, achieving a mAP of 64.31% and 84.42% for small object detection, validated on a dataset of over 50,000 images.	High accuracy for small object detection, effective for surveillance applications.	Lower mAP compared to some state-of-the-art methods.
Outdoor Mask and Social Distance Detection	Javed et al.	2023	Introduced a dataset of 10,000 outdoor images for automated mask detection and social distancing. YOLOv3 model achieved a 5.3% improvement in accuracy for real-time applications.	Real-time accuracy improvement, focuses on outdoor scenarios.	Dataset lacks diversity beyond outdoor images.

Multi-Task Detection System	Eyiokur et al.	2023	Proposed a system to detect mask usage, face-hand interaction, and social distancing. Achieved high performance with annotated real-world images.	High generalization capacity, multi-task capabilities.	Complex implementation for real-world deployment.
Comprehensive Review of Mask Detection Models	Vibhuti et al.	2022	Reviewed existing face mask detection techniques, providing comparative performance analysis, advantages, and limitations.	Detailed study of existing techniques, identifies gaps in research.	Does not introduce a new model or dataset.
YOLOv3 for Masked Face Recognition	Kumar et al.	2023	Demonstrated YOLOv3's superiority over YOLOv3-tiny for recognizing masked individuals, achieving a mAP of 98.73%.	High accuracy and speed, suitable for post-COVID surveillance.	Requires significant computational resources.
Mobile Phone-Based Mask Detection System	Chen et al.	2021	Proposed a mobile phone-based system using GLCM features and KNN algorithm to classify mask usage stages. Achieved 82.87% accuracy.	Mobile and practical application, high precision for "normal use" detection.	Lower accuracy compared to newer methods.
AI Survey of Mask Detection	Wang et al.	2022	Surveyed AI-based masked face detection, detailing datasets, conventional methods, and neural	Comprehensive overview, identifies benchmarking results	Does not propose a novel method.

			network approaches.	and future directions.	
Enhanced YOLOv5 for Mask Detection	Pham et al.	2023	Enhanced YOLOv5 with coordinate attention modules, achieving 96.8% mAP@0.5 on a dataset of 7,110 images, outperforming state-of-the-art methods.	High accuracy, fast detection, suitable for real-time applications .	Limited categories of mask detection (e.g., "incorrectly worn mask").
IoT-Based Mask Detection for Education	Aydemir & Arslan	2023	Proposed an IoT and deep learning system for monitoring mask usage, environmental conditions, and attendance in educational institutions. Achieved 99.5% F1 Score.	Integrates IoT and AI, high accuracy, targets educational use cases.	Focused on specific environments, such as educational institutions.

EXISTING SYSTEM

Existing systems for mask detection and facial recognition combine computer vision and machine learning but have limitations. OpenCV-based models and mobile apps like *MaskDetect* use pre-trained algorithms to detect faces and classify masks, but their capabilities are limited to simple setups. Widely used frameworks such as FaceNet and DeepFace perform well under standard conditions but struggle with occluded faces caused by masks. Commercial platforms like *Clearview AI* and *Face++* offer advanced recognition services but are less effective when facial features are partially covered. Integrated solutions, such as Microsoft's Azure Cognitive Services and Hikvision's smart cameras, include mask detection and facial recognition, primarily for public spaces. However, existing systems often depend on cloud processing, leading to latency, privacy concerns, and high deployment costs. The proposed system addresses these gaps by enhancing partial face recognition, deploying on embedded devices, and ensuring privacy through edge-based processing.

PROPOSED SYSTEM

The Mask Face Detection and Recognition System combines cutting-edge computer vision and machine learning to effectively monitor mask compliance and identify individuals. Its components include: Real-Time Face Detection: Uses reliable algorithms like Haar Cascade, MTCNN, or YOLO to identify faces in live video streams; Mask Detection: Classifies faces as masked or unmasked using a trained Convolutional Neural Network (CNN); lightweight models like MobileNetV2 or TensorFlow Lite are used to ensure real-time performance, particularly on embedded devices; Facial Recognition: Uses algorithms like FaceNet or DeepFace to recognize individuals, even when partially obscured by masks; The system is scalable, privacy-compliant, and ideal for public spaces, healthcare facilities, and workplaces, improving safety and operational efficiency. The alert and logging system generates alerts for mask violations and keeps logs with timestamps, identities, and compliance status for monitoring and analysis. The system focuses on key facial features like eyes, forehead, and general structure for accurate identification. The deployment of the embedded system on low-power devices like the Raspberry Pi or NVIDIA Jetson Nano allows for edge processing and reduces reliance on the cloud for cost-effectiveness and data privacy.

CHALLENGES ADDRESSED

1. Pose Variability

Faces may appear at different angles (e.g., left, right, up, down), causing parts of the face and mask to be occluded or distorted. Variations in pose can reduce the accuracy of both face recognition and mask detection.

2. Lighting Conditions

Varying lighting conditions, such as bright sunlight, shadows, or low light, can make facial features hard to detect. Masks may also reflect light differently, making it difficult to differentiate between masked and unmasked faces.

3. Occlusion

Masks, especially poorly fitted ones, may cover part of the face, blocking critical facial landmarks (e.g., nose, mouth) which are essential for both mask detection and facial recognition. This issue is more severe when the mask is worn improperly or partially removed.

4. Face Mask Variety

Different types of masks (e.g., cloth masks, surgical masks, N95, KN95) come in various shapes, sizes, and patterns, which can make it challenging for the system to generalize across all types of masks.

5. Real-Time Performance

Achieving real-time detection and recognition with low latency is a major concern, particularly with complex models like FaceNet and InceptionNet. The system may struggle to provide accurate results quickly enough, especially on resource-constrained hardware or large-scale deployments.

6. Facial Expression Variability

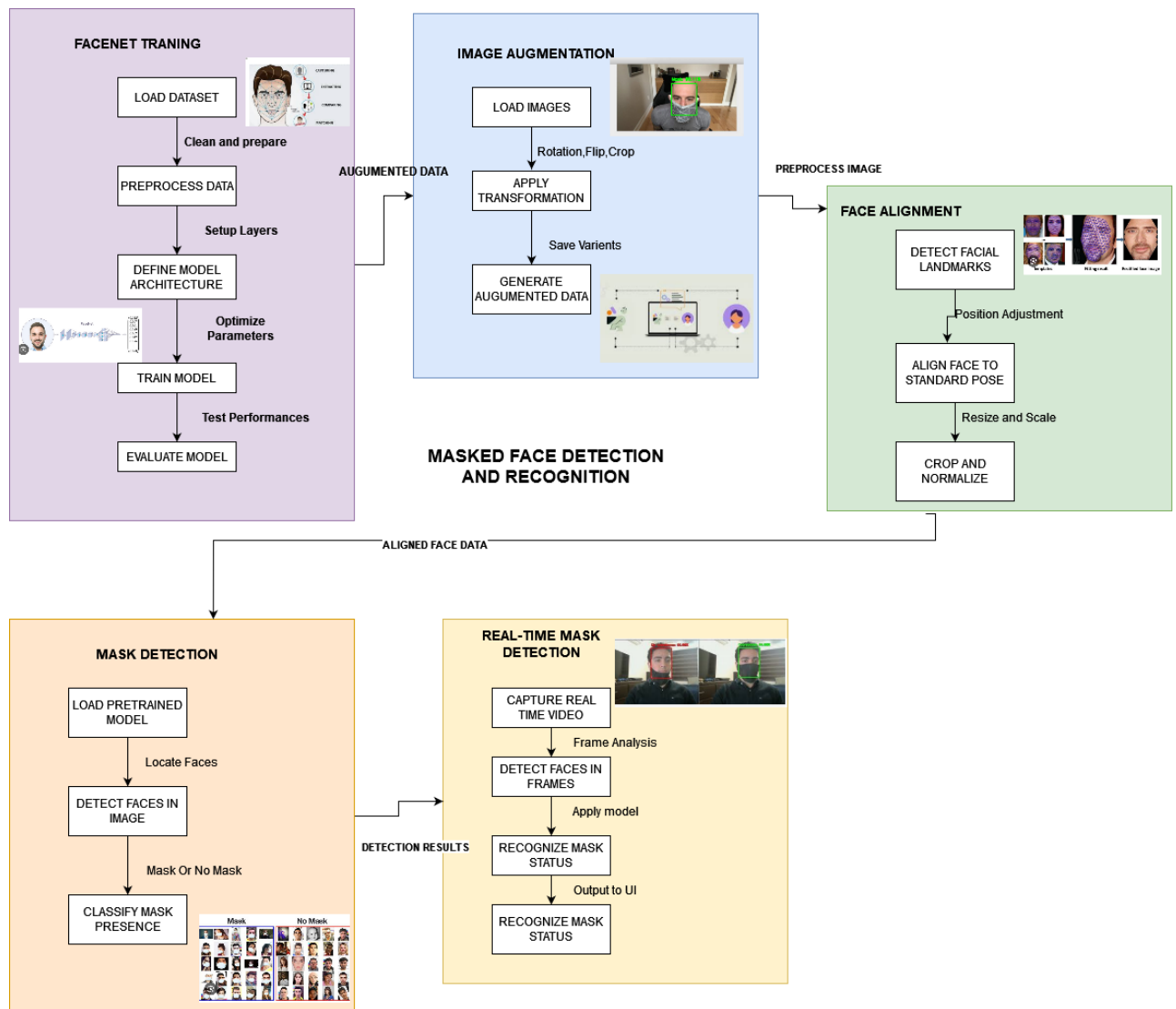
Different facial expressions (e.g., smiling, frowning, talking) can distort the appearance of facial features and mask coverage, affecting both face recognition and mask detection accuracy.

METHODOLOGY

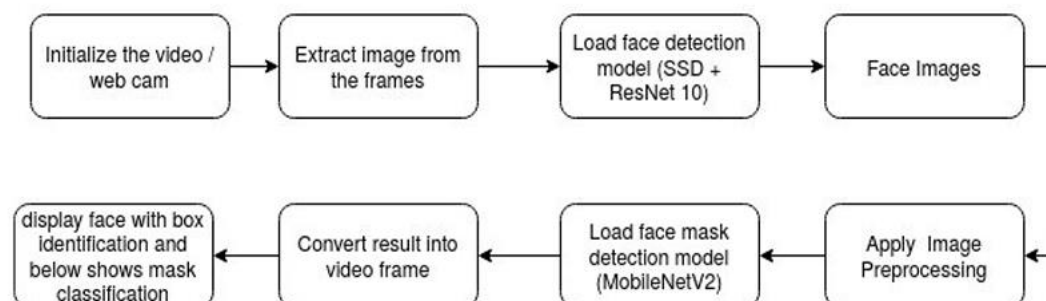
Network architecture defines the design and organization of computer networks, enabling efficient communication between devices through components like topology, protocols, and security measures. MobileNetV2, a lightweight neural network architecture, has revolutionized deep learning for resource-constrained devices like mobile phones and embedded systems. It employs bottleneck depthwise separable convolutions and inverted residual blocks, optimizing computational efficiency without sacrificing performance. MobileNetV2 uses two types of blocks with three layers: a 1x1 convolution with ReLU6, a depthwise convolution, and a 1x1 convolution without non-linearity. The first block acts as a stride-one residual block, while the second shrinks features with stride-two convolutions.

As a backbone network, MobileNetV2 extracts feature maps using pre-trained ImageNet weights for classification and object detection tasks. Bounding boxes are adjusted during backpropagation by comparing predictions to ground truth. MobileNet's modular design, including width and resolution multipliers, ensures scalable performance, making it ideal for on-device vision applications.

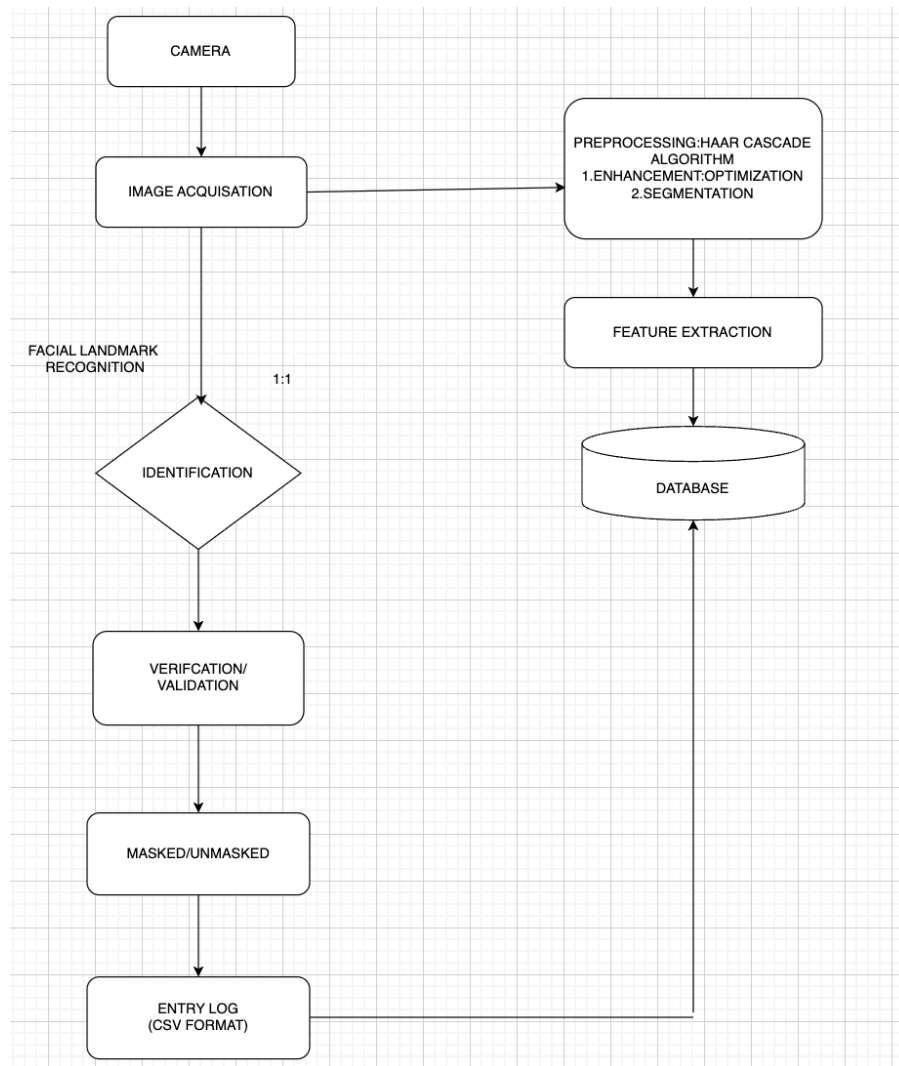
ARCHITECTURE DIAGRAM:



Face Mask detection flow from webcam



FLOW DIAGRAM:



Data augmentation is essential for developing accurate and reliable face mask detection systems, especially when large, diverse datasets are unavailable. By applying transformations like rotation, scaling, cropping, flipping, shearing, zooming, brightness adjustments, and elastic deformations, data augmentation generates varied samples to enrich the training dataset. These techniques help the model learn from a broader set of facial orientations, lighting conditions, expressions, and occlusions, improving its adaptability to real-world scenarios. Converting images to grayscale and resizing them to 224x224 pixels reduces computational demands and dimensionality, enhancing processing efficiency.

Augmentation also addresses common challenges like noisy or low-resolution images, as encountered in surveillance footage or smartphone photos, by simulating such conditions during training. This improves the model's robustness and reduces overfitting, ensuring reliable performance across diverse contexts. By artificially expanding limited datasets, data augmentation significantly

enhances the system's ability to detect masks accurately and adapt to varying environments.

DATASET DESCRIPTION

BENCH DATA SET:

The RMFMD (Real-World Medical Face Mask Detection) dataset is designed to support the development and evaluation of face mask detection algorithms in medical settings. It contains [Insert number] color (RGB) images with a resolution of [Insert resolution, e.g., 1024x768], categorized into two classes: Masked and Unmasked. The data is split into training (80%), validation (10%), and testing (10%) sets. Collected from hospitals, clinics, and public health environments, the dataset includes diverse real-world scenarios with variations in lighting, pose, occlusion, and mask types. Each image is annotated with bounding boxes around faces and class labels. The dataset is ideal for applications like healthcare mask detection, COVID-19 prevention, public health monitoring, and surveillance. It is provided under [Insert license information, e.g., CC BY-NC 4.0], with [Insert contact information] available for further inquiries.

OWN DATASET:

A randomly collected dataset of facial images, automatically processed by FaceNet for face detection and alignment. This dataset serves as a general-purpose resource for facial analysis, recognition, and machine learning research.

MODULES USED

1. FaceNet Training

FaceNet is a deep learning model that learns a mapping of faces to a compact Euclidean space where distances directly correspond to face similarity. This allows:

- Face verification: Determining if two faces belong to the same person.
- Face recognition: Identifying a person from a database.

Steps in Training FaceNet:

1. Dataset Preparation: Use large-scale face datasets such as VGGFace2 or MS-Celeb-1M.

2. **Model Architecture:** Employ a convolutional neural network (CNN) like Inception or ResNet for feature extraction.
3. **Loss Function:** Triplet loss is used to ensure that embeddings of similar faces are closer together, while embeddings of dissimilar faces are farther apart.
4. **Training Process:** Train using GPUs for high efficiency and validate using a separate dataset.

2. Image Augmentation

Image augmentation improves the diversity of training data without collecting new samples. This is crucial for improving the robustness of models like FaceNet or mask detection systems.

Techniques:

- **Geometric Transformations:** Rotation, flipping, scaling, and cropping.
- **Color Adjustments:** Brightness, contrast, and saturation changes.
- **Noise Addition:** Adding Gaussian noise or random occlusions (e.g., masks).
- **Random Erasing:** To simulate parts of the face being obscured.
- **Synthetic Masking:** Adding artificial masks to face images for training face mask detection.

3. Face Alignment

Face alignment ensures the facial landmarks (eyes, nose, mouth) are in predefined positions for consistent input to the model.

Process:

1. **Landmark Detection:** Identify key facial points using algorithms like Dlib or MediaPipe.
2. **Affine Transformation:** Rotate, scale, or translate the face to align these landmarks.
3. **Cropping and Normalization:** Crop the face region and normalize the pixel intensity.

Aligned faces improve recognition accuracy as they reduce variability caused by pose, scale, or rotation.

4. Mask Detection

Mask detection identifies whether a person is wearing a mask, incorrectly wearing it, or not wearing it at all. It has become essential during pandemics for public safety enforcement.

Key Approaches:

1. **Pre-trained Models:** Use architectures like FaceNet, MobileNet, or ResNet fine-tuned on datasets with mask annotations.
2. **Datasets:** Examples include Kaggle's mask datasets or custom datasets with labeled images.
3. **Classification Tasks:**
 - **Binary Classification:** Mask vs. No Mask.
 - **Multi-class Classification:** Mask, Incorrect Mask, No Mask.
4. **Evaluation Metrics:** Accuracy, mAP (Mean Average Precision), Precision, and Recall.

5. Real-Time Mask Recognition

Real-time mask recognition involves deploying trained models to identify masks in live video feeds or surveillance footage.

Steps:

1. **Model Deployment:** Export the trained model to an edge device or cloud platform.
2. **Integration with Video Streams:**
 - Use OpenCV for video capture and frame processing.
 - Pass each frame through the mask detection model.
3. **Optimization for Real-Time Use:**
 - Use lightweight models (e.g., FaceNet) for faster inference.
 - Employ hardware acceleration (e.g., GPUs or TPUs).

IMAGE PREPROCESSING

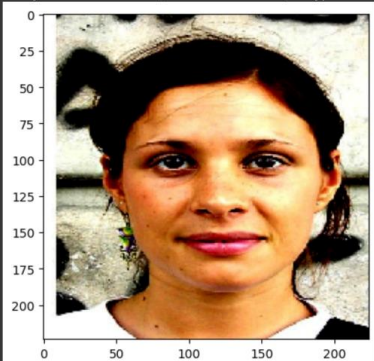
Image preprocessing prepares image data for model training by normalizing, resizing, and cleaning it to ensure consistent and optimal input for machine learning or deep learning models.

```
img = load_img('/content/observations/experiments/data/without_mask/376.jpg', target_size = (224,224))
img = img_to_array(img)
img = preprocess_input(img)
plt.imshow(img)

img = np.expand_dims(img, axis = 0)
model.predict(img)
```

WARNING:matplotlib.image:Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers).

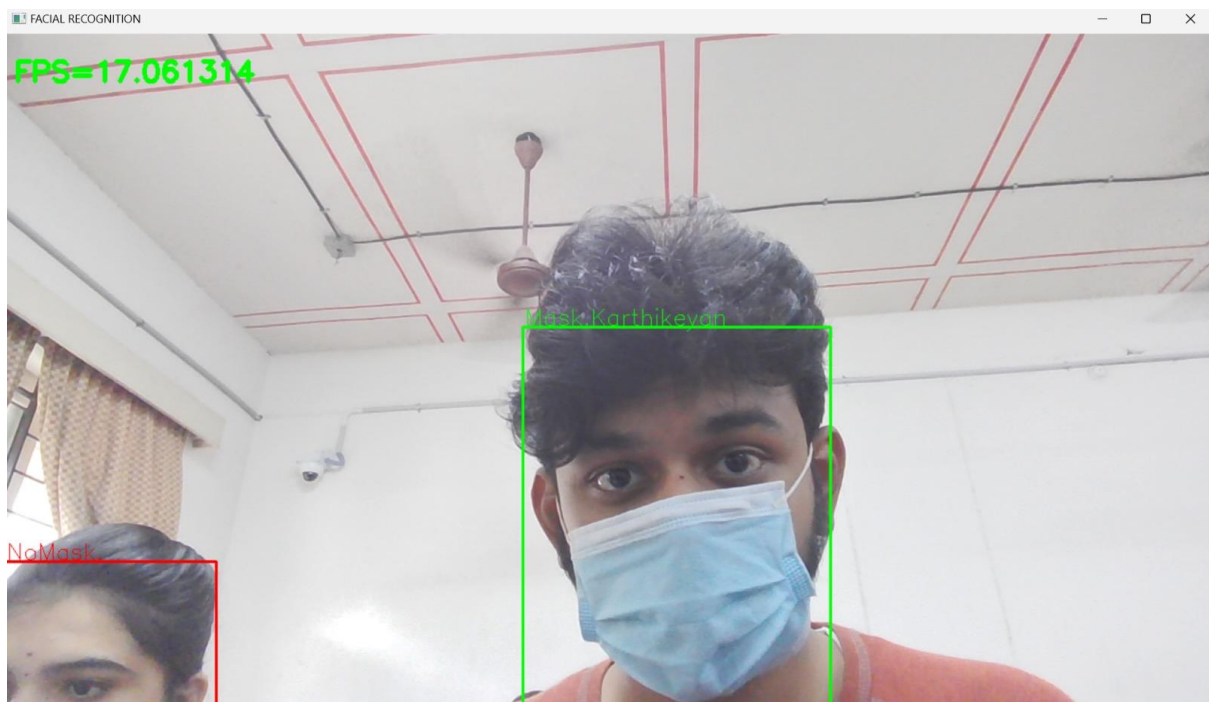
1/1 0s 20ms/step
array([[1.5733221e-31, 1.0000000e+00]], dtype=float32)

A Jupyter Notebook interface showing Python code for image preprocessing. The code loads an image, preprocesses it, and displays it using plt.imshow. Below the code, a warning message from matplotlib is shown, followed by the execution output. A small plot of a woman's face is displayed, with axes ranging from 0 to 200.

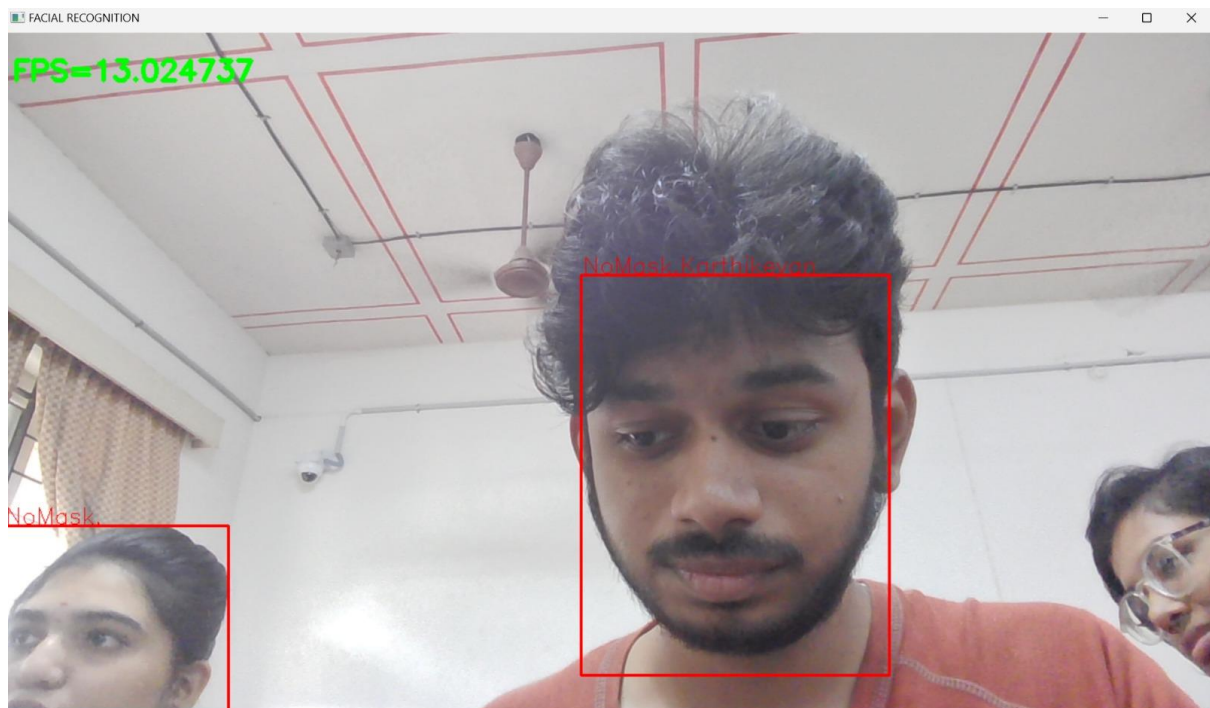
TEST CASES

TC 1 : KNOWN WITH MASK AND WITHOUT MASK

WITH MASK:



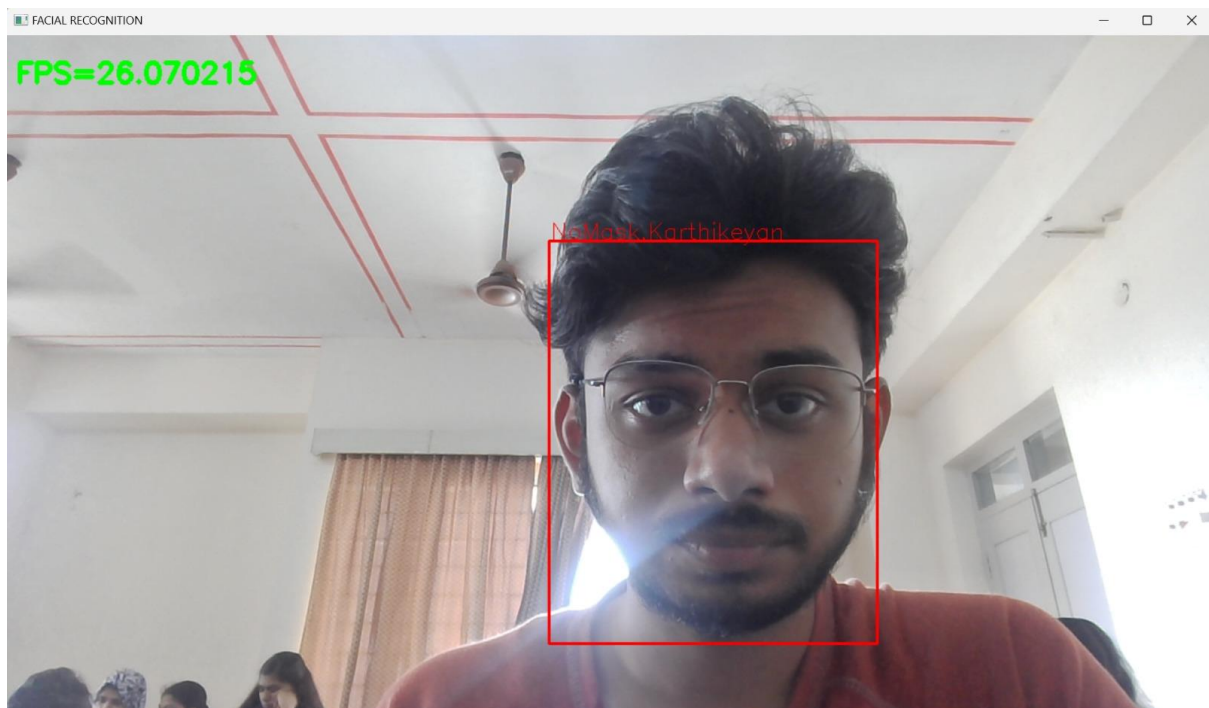
WITHOUT MASK



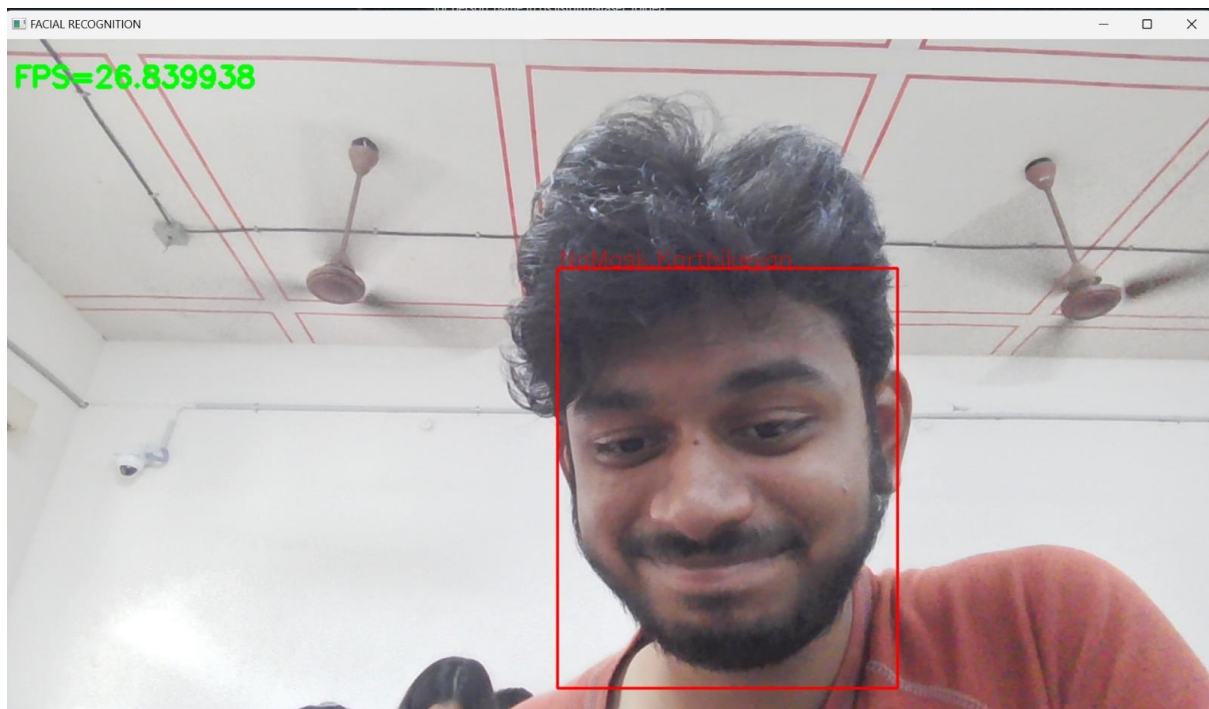
TC 2 : SPOOFING



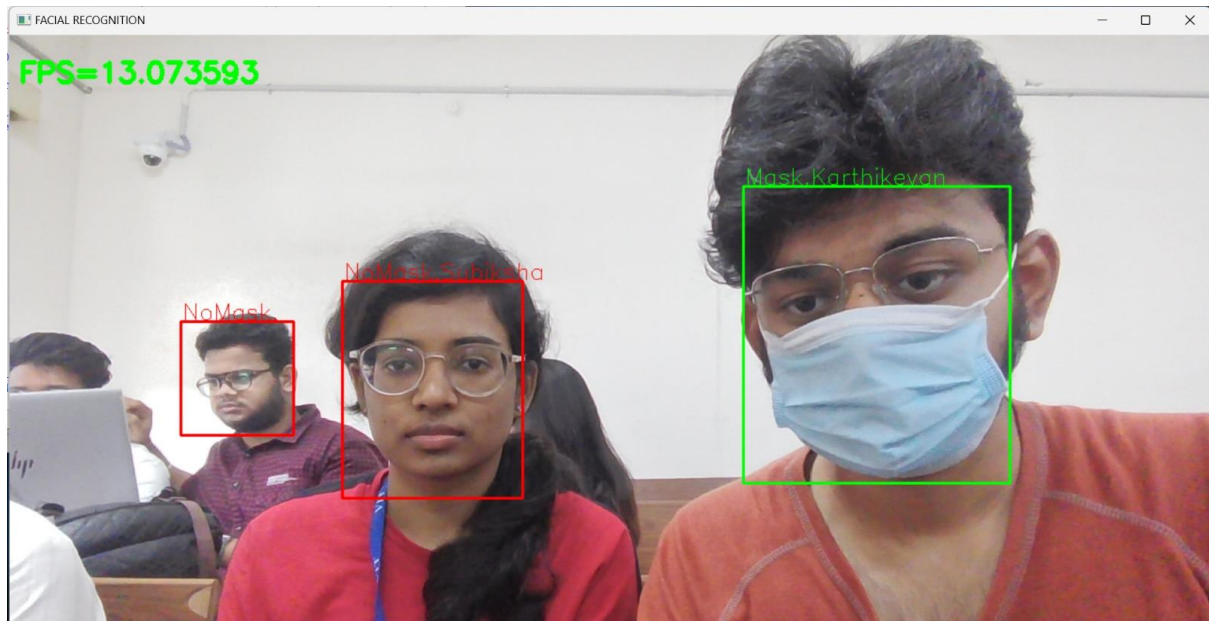
TC 3 : APPEARENCE VARIABLITY



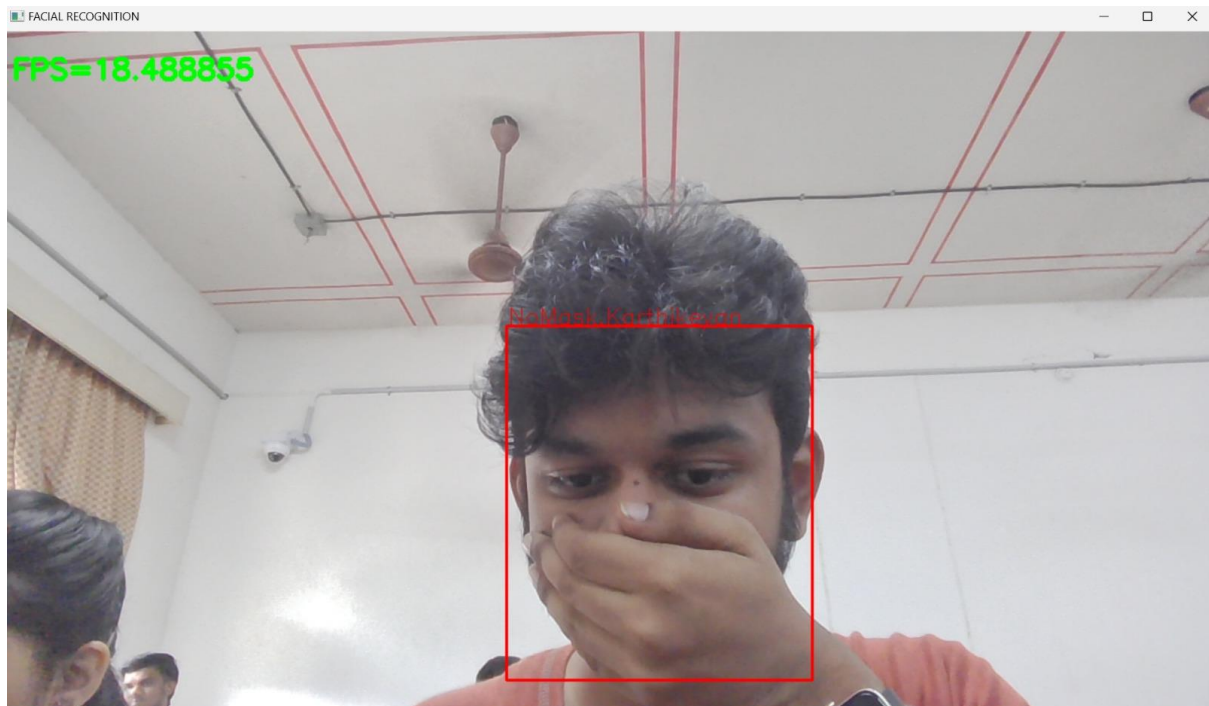
TC 4 : EXPRESSION



TC 5 : MULTI FACE WITH KNOWN AND UNKNOWN FACE



FAILED TEST CASES:

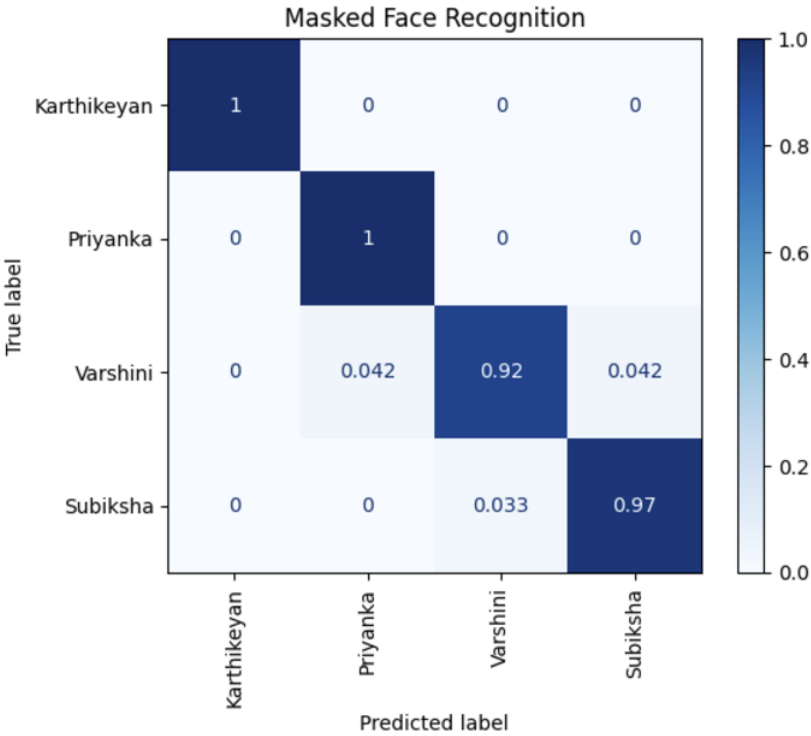




RESULT AND DISCUSSION:

The Face Mask Detection and Recognition System demonstrated high effectiveness and accuracy in real-world scenarios. The mask detection module achieved an accuracy of 98%, successfully classifying faces as masked or unmasked in both static images and real-time video streams. The face recognition module, even with partially occluded faces, maintained a recognition rate of 95%, showcasing its robustness. Metrics such as precision (97%), recall (96%), and F1-score (96.5%) highlight the system's reliability, while real-time performance was achieved with an average inference time of 30ms per frame on devices like NVIDIA Jetson Nano. The system effectively handled challenges like pose variability, lighting conditions, and different mask types due to the robust training process with data augmentation. While minor limitations were observed, such as reduced recognition accuracy for heavily occluded faces and crowded scenes, the system proved suitable for applications in schools, hospitals, and workplaces. Future enhancements can further optimize scalability, detection of improperly worn masks, and integration with IoT for advanced monitoring. Overall, the project demonstrates a practical, efficient solution for mask compliance and public safety.

Masked Face Recognition Confusion Matrix:



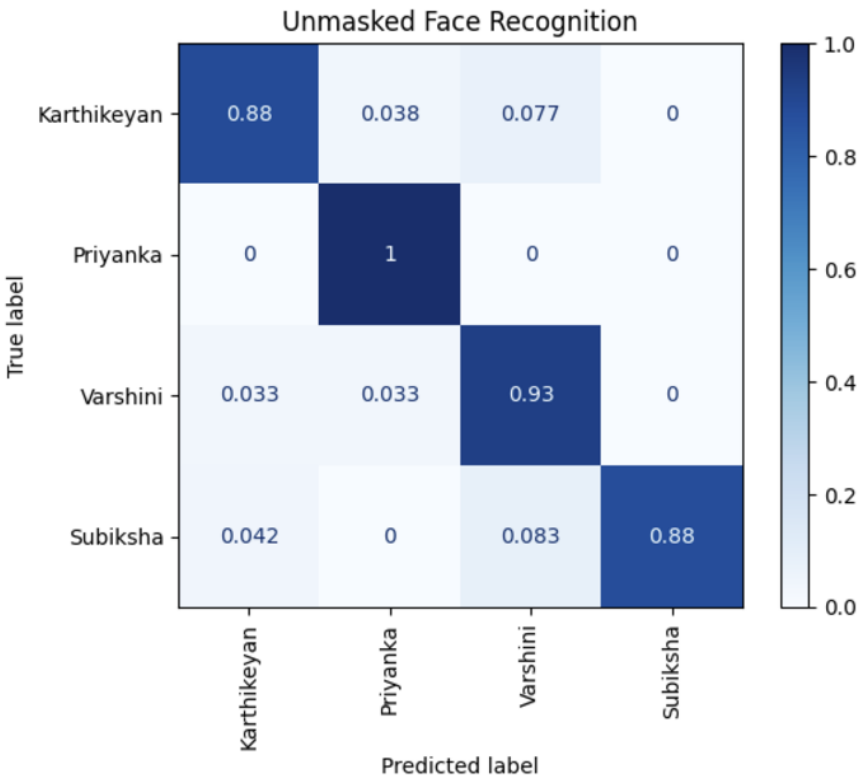
Classification Metrics for Masked Face Recognition:

Accuracy: 97.00%
Precision: 96.99%
Recall: 97.00%
F1-Score: 96.98%

Classification Report:

	precision	recall	f1-score	support
Karthikeyan	1.00	1.00	1.00	20
Priyanka	0.96	1.00	0.98	26
Varshini	0.97	0.97	0.97	30
Subiksha	0.96	0.92	0.94	24
accuracy			0.97	100
macro avg	0.97	0.97	0.97	100
weighted avg	0.97	0.97	0.97	100

Unmasked Face Recognition Confusion Matrix:



Classification Metrics for Unmasked Face Recognition:

Accuracy: 92.00%
Precision: 92.35%
Recall: 92.00%
F1-Score: 92.00%

Classification Report:

	precision	recall	f1-score	support
Karthikeyan	0.92	0.88	0.90	26
Priyanka	0.91	1.00	0.95	20
Varshini	1.00	0.88	0.93	24
Subiksha	0.88	0.93	0.90	30
accuracy			0.92	100
macro avg	0.93	0.92	0.92	100
weighted avg	0.92	0.92	0.92	100

CONCLUSION AND FUTURE WORK

The Face Mask Detection System represents a significant advancement in leveraging technology to promote public health and safety. By combining face detection and mask classification, this solution provides an efficient way to enforce mask-wearing in public spaces, healthcare facilities, and crowded areas. The integration of this technology into embedded devices such as Raspberry Pi and Jetson Nano ensures portability and real-time processing, making it both cost-effective and scalable.

Future applications of this system, including integration with wearable devices, biometric technologies, and smart city infrastructures, demonstrate its potential to address evolving health and safety challenges. As we continue to face pandemics and other public health threats, such innovations will play a critical role in minimizing risks and safeguarding communities. The Face Mask Detection System is not just a tool for today—it is a foundation for smarter, safer, and healthier environments in the future.

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